Description

Illustration

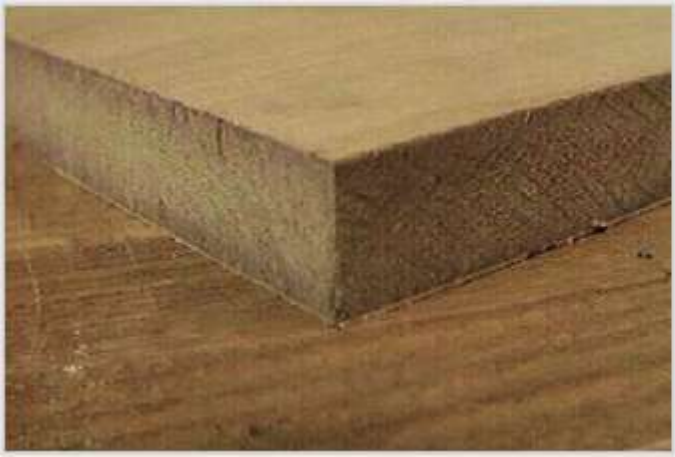




Figure caption

1. Cross section of an MDF board. © ANSYS, Inc. 2. MDF being hand-milled to create a decorative partition wall. Image used under license from Shutterstock.com

The material

Fiberboard is made in the same way as hardboard: by hot (140 - 200°C) compression of wood fibers from which the carbohydrates and other soluble constituents have been leached out, increasing the proportion of lignin, which acts as a binder. The resulting sheet products have smooth surfaces comparable with that of wood. They are hard, stiff and, with the addition of a little urea-formaldehyde or phenol-formaldehyde binder, water resistant. Most have a density in the range 600 - 800 kg/m3. The record describes MDF, but there are many other grades. MDF provides a superior surface for high tech finishes including lacquers, foils, thin paper laminates and other popular coatings. It has excellent machinability giving clean, sharp edges in cutting and drilling operations and it takes screws and staples well. They include fire resistant (FR) grades than meet Class 1 fire spread standards, water resistant grades for external use, and formaldehyde-free grades to minimize emissions.

Compositional summary

Cellulose/Hemicellulose/Lignin/12%H2O/Adhesive

General properties

Density		600	-	900	kg/m^3
Price	*	0.302	-	0.34	GBP/kg

Material form that data applies to

Bulk	✓
Sheet	✓

Building system

Superstructure	✓
Enclosure	✓

Interiors	✓			
Services	✗			

Mechanical properties

Young's modulus	* 1.8	-	3.2	GPa
Shear modulus	* 0.7	-	0.8	GPa
Bulk modulus	* 1.7	-	2.9	GPa
Bending modulus	2	-	3.5	GPa
Poisson's ratio	* 0.2	-	0.3	
Yield strength (elastic limit)	* 4.5	-	5.5	MPa
Tensile strength	8.5	-	21	MPa
Compressive strength	3	-	7	MPa
Bending strength	28	-	30	MPa
Elongation	* 0.3	-	0.36	% strain
Hardness - Vickers	* 6.1	-	7.5	HV
Fatigue strength at 10^7 cycles	* 7.4	-	10.3	MPa
Fracture toughness	* 1	-	2	MPa.m^0.5
Mechanical loss coefficient (tan delta)	* 0.013	-	0.02	

Thermal and combustion properties

Thermal conductor or insulator?	Good insulator			
Thermal resistivity	7.14	-	8.33	m.°C/W
Thermal expansion coefficient	* 10	-	15	µstrain/°C
Specific heat capacity	1.66e3	-	1.71e3	J/kg.°C
Maximum service temperature	120	-	140	°C
Minimum service temperature	-73	-	-23	°C
Flammability	Highly flammable			
Emissivity	0.81	-	0.9	

Hygro-thermal properties

Water absorption	12	-	35	%
Water vapor permeability	2.94e-11	-	7.27e-11	kg.m/s.m^2.Pa
Frost resistance	Good			

Electrical properties

Electrical conductor or insulator?	Poor insulator			
Electrical resistivity	* 1.05e13	-	2.45e15	µohm.cm
Dielectric constant (relative permittivity)	* 6	-	30	
Dissipation factor (dielectric loss tangent)	* 0.3	-	0.4	

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

Dielectric strength (dielectric breakdown)	* 1	-	2	MV/m

Optical properties

Natural color

Light brown

Transparency	Opaque	
Transmissivity	0	%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Average
Sound isolation	Average

Processability (scale 1 = impractical to 5 = excellent)

Moldability	3	-	4
Machinability	5		

Durability

Water (fresh)	Limited use
Water (salt)	Limited use
Weak acids	Limited use
Strong acids	Unacceptable
Weak alkalis	Acceptable
Strong alkalis	Unacceptable
Organic solvents	Acceptable
UV radiation (sunlight)	Good
Wear resistance	Unacceptable
Industrial atmosphere	Limited use
Rural atmosphere	Limited use
Marine atmosphere	Limited use

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 0.866	-	0.955	kg/kg
Embodied energy (virgin grade production)	* 19.4	-	21.4	MJ/kg
Water Usage	* 665	-	735	l/kg

Material processing: energy

Machining, coarse, embodied energy	* 14.9	-	16.6	MJ/kg
Machining, fine, embodied energy	* 15.5	-	17.6	MJ/kg
Machining, grinding, embodied energy	* 16.1	-	18.6	MJ/kg

Material processing: climate change

Machining, coarse, climate change (CO2-eq)	* 0.974	-	1.09	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.01	-	1.15	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.05	-	1.22	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.45	-	0.497	kg/kg
Embodied energy, recycling	* 16.1	-	17.8	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 19.7	-	20.7	MJ/kg
Combustion CO2	* 1.73	-	1.82	kg/kg
Landfill	✓			
Biodegrade	✓			
A renewable resource?	✓			

Supporting information

Typical uses

In the furniture industry: cabinet carcasses, drawers, doors, unit tops, particularly where profiled edges, relief surfaces and one piece framing is required. In substitution for solid timber: machining of consistent straight lengths of moldings, shop-fitting and partitioning systems. The smoothness of MDF surfaces makes them ideal for painting and the application of thin paper, veneers or PVC foils. Moisture resistant MDF in conjunction with an appropriate coating system is intended for external use such as business signs, shop-fronts and fascias, exterior display stands and marine craft interiors.

Links

Reference